CY110

FINAL EXAMINATION

SEPTEMBER 2022

There are SIX questions in this paper

Candidates MUST answer ANY FIVE QUESTIONS in this PAPER

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risk ass measure

QUESTION 1

Explain what is meant by a risk assessment in the context of network security and provide FOUR (4) reasons why it is important.

10 Marks

What are FOUR commonly used decisions that depend on management's attitude relating to identified and prioritized risks

10 Marks

the extent of the the likelihood an entity is threaten and the likelihoh

Total 20 Marks

QUESTION 2

Alice and Bob are having a debate about Public Key Infrastructure (PKI). Bob says that it is simply a way of authenticating users. However, Alice argues that it is a type of encryption algorithm. They have asked you to decide who is correct.

1 Mark

a) Briefly outline the purpose of PKI. You should also explain what is meant by certificate authority; digital certificate and digital signatures

9 Marks

b) Explain cryptographic protocols, phases and details involved in **SSL** securing communication in a case where JAMES wishes to access transactions made between him and Kapasa Makasa University through the Bank's URL: www.stanbic.co.zm

10 Marks

PKI is used to provid reference p verification or valid

Total 20 Marks

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QUESTION 3

a) Explain and clearly distinguish the terms providing good examples where necessary

(i) Vulnerability
(ii) Threat
(iii) Risk

12 Marks

b) Identify any two common types of attacks and provide best methods of Mitigation or treatment

8 Marks

Risk is assesment of a threat. Potential occurrance and event

Total 20 Marks

QUESTION 4

In symmetric encryption **VPNs**, IPsec is a suite of protocols running on network devices for securing the networks involved in the communication

a) Explain how end to end devices achieve secure encryption, authentication, outlining cryptographic protocols involved and phases where necessary

10 Marks

b) Explain also how these encryption protocols provide confidentiality and integrity of the payload moving in the VPN network communication channel

10 Marks

Total 20 Marks

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QUESTION 5

(a) Briefly describe the processes of encryption and decryption in relation to Cryptography

10 Marks

(b) What is the difference between Symmetric and Asymmetric key encryptions ?

10 Marks

Total 20 Marks

QUESTION 6

When users have stated their identity, it is necessary to validate that they are the rightful owners of that identity

a) Define the term Authentication **2 Marks**

b) Explain with examples stronger methods of:
   (i) Single factor Authentication (SFA)
   (ii) Multi factor Authentication (MFA) **8 Marks**

c) In the context of cyber security threat sources arise from a combination of threat-actors. Identify 5 typical threat-actors facing organization in cybersecurity **10 Marks**

Total 20 Marks